



Roll No:

BCA
(SEM. II) THEORY EXAMINATION 2025-26
MATHEMATICAL FOUNDATION

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define Symmetric Matrix with example	CO-1	K1
b.	Examine for linear independence or dependence of vectors: $(2, -1, 3), (8, 2, 0), (0, 1, -2)$	CO-2	K1
c.	Find the value of $\frac{d(x^2 + 4x + e^x)}{dx}$	CO-3	K1
d.	Solve $(D^2 - 3D + 2)y = 0$	CO-4	K1
e.	Evaluate of $L(e^{-t} \cos t)$	CO-5	K1
f.	Define Graph.	CO-1	K1
g.	Evaluate $\int (t^3 + 2t - 6) dt$	CO-2	K1

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Q no.	Question	CO	Level
a.	Show that $\begin{vmatrix} b+c & a+b & a \\ c+a & b+c & b \\ a+b & c+a & c \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$	CO-1	K3
b.	Examine the consistency of the system and if consistent, solve the equation: $x + y + 2z = 9, 2x + 4y - 3z = 1, 3x + 6y - 5z = 0$	CO-2	K3
c.	Find the n^{th} order derivative of $\frac{1}{(x-3)}$	CO-3	K3
d.	Solve $(D^2 - 4)y = e^{2x} + e^{-4x}$	CO-4	K3
e.	Find the Laplace Transform of $\int_0^t e^{-2t} \sin t dt$	CO-5	K3

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
	Using row elementary operation, find the inverse of the matrix A, where	CO-1	K3



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	$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$		
b.	Reduce the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$ to echelon form and, hence, find its rank.	CO-1	K3

4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Determine the eigen values and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$	CO-2	K3
b.	Verify the Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$	CO-2	K3

5. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$	CO-3	K3
b.	If $u = xyz$, $v = x^2 + y^2 + z^2$, $w = x + y + z$, find the Jacobian $\frac{\partial(u, v, w)}{\partial(x, y, z)}$	CO-3	K3

6. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Evaluate $\int x^2 \sin x dx$	CO-4	K3
b.	To show that $\beta(m, n) = \beta(n, m)$	CO-4	K3

7. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Use convolution theorem to evaluate: $L^{-1} \left\{ \frac{s}{(s^2 + 4)^2} \right\}$	CO-5	K3
b.	Define (I) Euler graph (II) Hamiltonian graph	CO-5	K3